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A PROJECT REPORT ON

"ADAPTIVE AI-BASED PLAGIARISM DETECTION SYSTEM"

A Dissertation Submitted in partial fulfilment of the requirement for the degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE ENGINEERING

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I. INTRODUCTION

Plagiarism has moved beyond copied essays. Students and job seekers now reuse text in reports, resumes, portfolios, GitHub project descriptions and viva answers, often rewording it to avoid detection. Conventional checkers rely mainly on exact string matching and report a single similarity percentage, which says little about how the content was copied.

Multiple techniques can detect different forms of copying. Exact matching catches verbatim reuse, TF-IDF with cosine similarity captures shared vocabulary, n-gram overlap exposes lightly edited phrases, and transformer-based embeddings such as Sentence-BERT detect paraphrased text with the same meaning. Each technique is strong on one plagiarism type and weak on others.

However, most existing tools apply one fixed method, or a fixed blend of methods, to every document and give little explanation for their score.

The Adaptive AI-Based Plagiarism Detection System aims to design and implement an intelligent, end-to-end detection system that integrates:

- Multi-technique similarity analysis (exact match, TF-IDF, n-gram, Sentence-BERT)
- Plagiarism-type classification from the similarity pattern
- Dynamic adaptation of detector weights to the identified type
- Explainable results with visual analytics and PDF reports
- Web-based deployment through a Streamlit interface

This integrated approach ensures not only a more accurate similarity score but also an interpretable, practical tool for academic and professional use.

II. MOTIVATION BEHIND THE PROJECT WORK

Although plagiarism checkers are widely available, existing systems face key limitations:

- A single similarity percentage with no indication of the type of plagiarism
- One fixed detection method or fixed weights for every kind of content
- Weak detection of paraphrased and semantically copied text
- Little explanation of why a document was flagged

- Limited support for resumes, portfolios, GitHub projects and viva material

This motivates the need for a more intelligent and transparent solution that goes beyond a raw score and provides:

- Multi-technique, layered similarity analysis
- Identification of the likely plagiarism type
- Detector weights that adapt to that type
- Explainable results and reports
- A usable prototype for students and evaluators

The Adaptive AI-Based Plagiarism Detection System addresses these limitations by combining multiple similarity techniques, type classification, adaptive weighting and explainability in one framework.

III. PROBLEM STATEMENT DEFINITION

Existing plagiarism detection systems predominantly rely on a single similarity technique, or a fixed combination of techniques, and return one percentage score for the whole document. Such a score cannot show whether the content was copied verbatim, lightly reworded or paraphrased, and it gives evaluators no clear reasoning to act on.

Moreover, most tools are built for long-form academic text and are not tuned for resumes, portfolios, GitHub project descriptions or viva answers, where plagiarism patterns differ.

The problem addressed in this project is to develop an intelligent, adaptive detection system that:

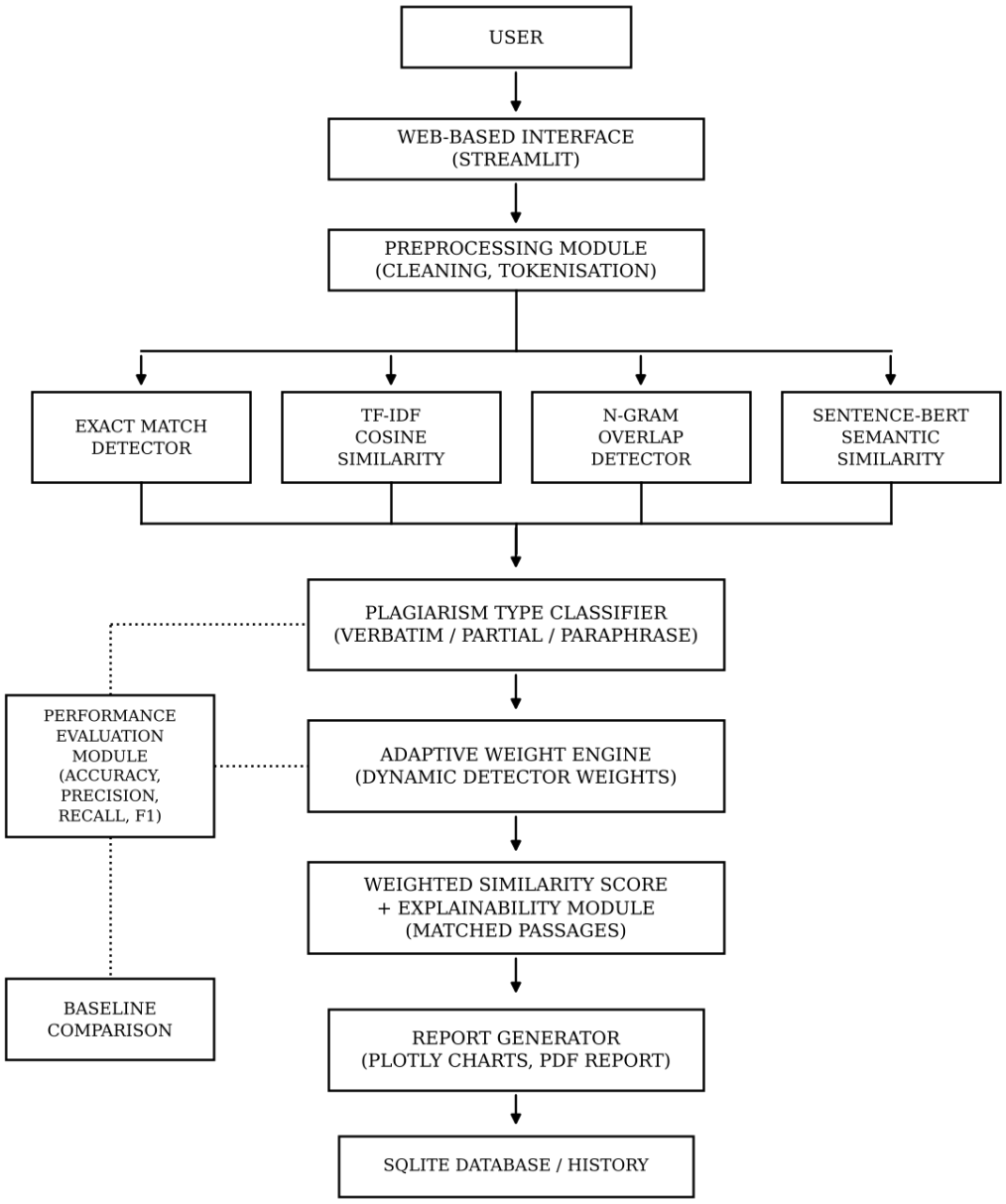
- Measures similarity using multiple complementary techniques
- Classifies the likely type of plagiarism from the similarity pattern
- Dynamically adjusts detector weights based on that type
- Provides explainable results through visual outputs and reports
- Enables easy use through a web-based interface

Such a system must improve detection accuracy, transparency and practical usability beyond conventional single-score checkers.

IV. STATEMENT OF OBJECTIVES

- To implement multiple similarity techniques (exact match, TF-IDF, n-gram and Sentence-BERT semantic similarity) for comparing submitted content against reference sources.
- To design a classifier that identifies the likely plagiarism type (verbatim copy, partial reuse or paraphrasing) from the combined similarity pattern.
- To develop an adaptive weighting mechanism that dynamically adjusts each detector's contribution based on the identified plagiarism type.
- To generate explainable results, including matched passages, per-technique scores and visual analytics using Plotly/Matplotlib.
- To build a Streamlit-based web prototype with SQLite storage that supports content upload, analysis and history, and produces PDF reports using ReportLab.
- To evaluate system performance using accuracy, precision, recall and F1-score.
- To compare the adaptive system against fixed-weight and single-technique detectors and demonstrate improved detection and interpretability.

V. SYSTEM ARCHITECTURE



VI. REFERENCES

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